Task 14: Design a simple microservices-based system

Aim

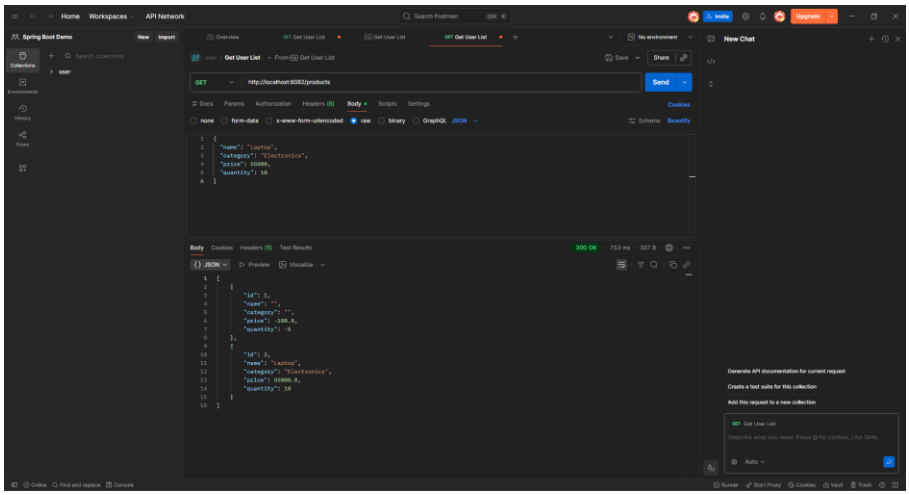
To design a simple **microservices-based system** by splitting a monolithic application into at least **two independent services**, and to apply microservices principles such as **loose coupling**, **single responsibility**, and **independent deployment**.

Algorithm

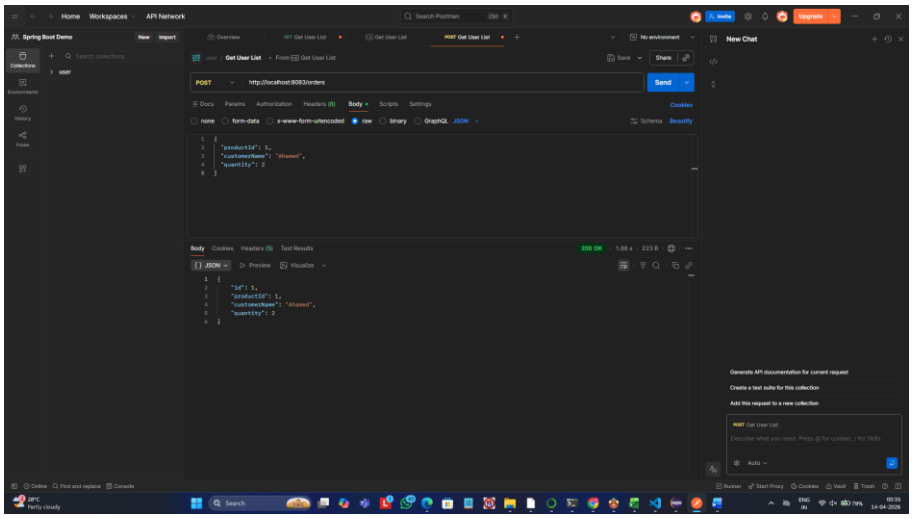
1. Start the system design process.
2. Identify modules in the monolithic application.
3. Separate the application into at least two independent services.
4. Assign one clear responsibility to each service.
5. Design each service with its own database and business logic.
6. Establish communication between services using REST APIs.
7. Ensure services are loosely coupled.
8. Allow each service to be developed, deployed, and scaled independently.
9. Test the interaction between services.
10. Stop the process.

Output:

Product services:



Order Services:



Result:

To design a simple **microservices-based system** by splitting a monolithic application into at least **two independent services**, and to apply microservices principles such as **loose coupling**, **single responsibility**, and **independent deployment** is executed successfully.